Hypothesis

Having daughters leads to stronger support for gender equality (as measured by attitudes).
That is, denoting X as a measure of support for gender equality,
(X for respondents with daughters) > (X for respondents without daughters).

1. Variables

- **Dependent Variable (Outcome)**

We construct a gender attitudes index as follows:

variable	statement	dataset		cohorts	valid obs	recoding
plj0625	At home, the husband should have the final say.	pl	bgp/bgpr389 bhp/bhp_508_q56 bip/bip_471_q107	2016 2017 2018	4166 2827 420	only consider values from 1 to 7
plj0624	For parents, vocational training or higher education for their sons should be more important than vocational training or higher education for their daughters.	pl	bgp/bgpr388 bhp/bhp_507_q56 bip/bip_470_q107	2016 2017 2018	4117 2803 416	only consider values from 1 to 7
plj0621	Having a job is the best way for a woman to be independent.	pl	bgp/bgpr385 bhp/bhp_504_q56 bip/bip_467_q107	2016 2017 2018	4105 2793 415	only consider values from 1 to 7
plj0622	Even a married woman should have a paid job so that she can be financially independent.	pl	bgp/bgpr386 bhp/bhp_505_q56 bip/bip_468_q107	2016 2017 2018	4099 2796 417	only consider values from 1 to 7

plj0623	If a woman earns more money than her partner, this inevitably leads to problems.	pl	bgp/bgpr387 bhp/bhp_506_q56 bip/bip_469_q107	2016 2017 2018	3983 2745 406	only consider values from 1 to 7

Additional coding and transformation
To ensure consistency across all survey items, we standardize the direction of responses so that higher values always indicate stronger support for gender equality.

Construction of the Standardized Average Index
Let x_ij be the recoded response of respondent i to item j. Let μ_j and σ_j be the mean and standard deviation of item j across all respondents.

1. Standardization:
For each item j and year, compute the z-score:
$z_{ij} = (x_{ij} - \mu_j) / \sigma_j$

2. Index construction:
Compute the mean of the standardized responses for each individual:
$X_i = (1 / n_i) * \sum z_{ij}$ over j in J_i, where J_i is the set of items answered by respondent i.

3. Interpretation:
- The index X_i has mean 0 and is in standard deviation units.
- Higher values indicate more egalitarian attitudes.

- **Independent Variable: Having Daughters**

The number of sons and daughters can be identified using the variables kidsex01-kidsex19 in the biobirth dataset of the SOEP.

These variables record the biological sex of each child, coded as 1 = male and 2 = female. By counting the occurrences of each value for a given respondent or household, one can determine the number of sons, daughters, and kids respectively.

As an explanatory variable, we define "At Least 1 Girl" as a binary indicator that equals 1 if an individual has at least one daughter, and 0 if the individual has exclusively sons.

- **Control Variables**

We include a set of control variables to account for potential confounders:

Concept	Variable Name (Example)	Dataset	Why It's a Key Control	valid obs	recoding

Age	bioage	biobirth	Older respondents are more likely to have children and may hold more traditional views.	12398	only consider positive values, take ln(x)
Dummy for gender (respondent)	sex	biobirth	Gender affects both attitudes and fertility patterns.	24317	1 if sex=2 (female), 0 otherwise
Dummys for the number of children	kidsex01-kidsex19	biobirth	More children → higher chance of having a daughter; family values may vary by family size.	≤9862	19 dummy variables, 1 if the number of children matches, otherwise 0
Dummys for martial status	pgfamstd	pgen	Being married or in a partnership may influence both the likelihood of having children (including daughters) and one's social and political attitudes, including views on gender equality.	74742	4 dummy variables for "married", "married but separated", "divorced", "widowed", 1 if applicable, 0 otherwise
Current net labor income	pglabnet	pgen	Net labor income is an essential control variable as income levels can shape access to education, social networks, and exposure to egalitarian values, all of which may affect attitudes toward gender equality.	39877	only consider positive values, take ln(x)
Dummy for migration background	migback	ppathl	Cultural background can influence both fertility and gender role attitudes.	129787	1 if migback=2 (direct migration background) or 3 (indirect migration background), 0 otherwise
Year of observation	syear	pl	Year fixed effects		Year fixed effects
Never Changing Person ID	pid	pl	Individual fixed effects		Individual fixed effects

2. Inclusion/Exclusion Criteria

Include:

- Respondents who participated in the individual questionnaire (pl) in any year from 2016 to 2018.

Exclude:

- Respondents without any biological children
- Respondents with fewer than 2 answered gender attitude variables
- Respondents with missing values for the key controls
- Respondents with unmatched IDs (pid) across datasets

3. Exact analysis

Panel Approach (with Fixed Effects): OLS regression with robust standard errors (clustered at person level for panel specifications). Panel model:

$X_it = \beta_0 + \beta_1 \cdot$ *At Least 1 Girl*$_it + \gamma' Z_it + \alpha_i + \delta_t + \varepsilon_it$

Where:

- X = gender attitudes index
- "At Least 1 Girl" = indicator from biobirth
- Z = controls from biobirth, pgen, ppathl
- δ_t = year fixed effects
- α_i = person fixed effects (if panel)

4. Interpretation of Results

- Focus: Sign and statistical significance of β_1
- Report:
  - Coefficient estimates
  - Robust standard errors
  - t-values and p-values
  - 95% confidence intervals
- Hypothesis is supported if $\beta_1 > 0$ and statistically significant ($p < 0.05$)

Pre-Analysis Plan: H2

Hypothesis

Having daughters leads to more left-leaning political preferences.
That is, denoting Y as a measure of left-leaning political preferences,
(Y for respondents with daughters) > (Y for respondents without daughters).

1. Variables

- **Dependent Variable (Outcome)**

We construct a left-leaning political preferences index as follows:

variable	statement	dataset		cohorts	valid obs	recoding
plh0004	Political attitude	pl	vp/vp133 zp/zp126 bep/bep122 bjp/bjp_171	2005 2009 2014 2019	20286 19965 25903 24818	only consider values from 0 to 10

Additional coding and transformation
We standardize the direction of responses so that higher values indicate stronger left-leaning political preferences.

Additionally, in each year, we compute the z-score:
$z_{ij} = (x_{ij} - \mu_j) / \sigma_j$

Interpretation:
- The index X_i has mean 0 and is in standard deviation units.
- Higher values indicate higher left-leaning political preferences.

- **Independent Variable: Having Daughters**

The number of sons and daughters can be identified using the variables kidsex01-kidsex19 in the biobirth dataset of the SOEP.

These variables record the biological sex of each child, coded as 1 = male and 2 = female. By counting the occurrences of each value for a given respondent or household, one can determine the number of sons, daughters, and kids respectively.

As an explanatory variable, we define "At Least 1 Girl" as a binary indicator that equals 1 if an individual has at least one daughter, and 0 if the individual has exclusively sons.

- **Control Variables**

We include a set of control variables to account for potential confounders:

Concept	Variable Name (Example)	Dataset	valid obs	recoding
Age	bioage	biobirth	12398	only consider positive values, take ln(x)
Dummy for gender (respondent)	sex	biobirth	24317	1 if sex=2 (female), 0 otherwise
Dummys for the number of children	kidsex01-kidsex19	biobirth	≤9862	19 dummy variables, 1 if the number of children matches, otherwise 0
Dummys for martial status	pgfamstd	pgen	74742	4 dummy variables for “married”, “married but separated”, “divorced”, “widowed”, 1 if applicable, 0 otherwise
Current net labor income	pglabnet	pgen	39877	only consider positive values, take ln(x)
Dummy for migration background	migback	ppathl	129787	1 if migback=2 (direct migration background) or 3 (indirect migration background), 0 otherwise
Year of observation	syear	pl		Year fixed effects
Never Changing Person ID	pid	pl		Individual fixed effects

2. Inclusion/Exclusion Criteria

Include:

- Respondents who participated in the individual questionnaire (pl) in 2005, 2009, 2014, or 2019

Exclude:

- Respondents without any biological children
- Respondents with no recorded political preference variable
- Respondents with missing values for the key controls
- Respondents with unmatched IDs (pid) across datasets

3. Exact analysis

Panel Approach (with Fixed Effects): OLS regression with robust standard errors (clustered at person level for panel specifications). Panel model:

$X_it = \beta_0 + \beta_1 \cdot$ At Least 1 Girl_it + $\gamma' Z_it + \alpha_i + \delta_t + \varepsilon_it$

Where:

- X = left-leaning political preferences index
- “At Least 1 Girl” = indicator from biobirth
- Z = controls from biobirth, pgen, ppathl
- δ_t = year fixed effects
- α_i = person fixed effects (if panel)

4. Interpretation of Results

- Focus: Sign and statistical significance of β_1
- Report:
  - Coefficient estimates
  - Robust standard errors
  - t-values and p-values
  - 95% confidence intervals
- Hypothesis is supported if $\beta_1 > 0$ and statistically significant ($p < 0.05$)

Pre-Analysis Plan: H3

Hypothesis

Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

That is, denoting Z as a measure of gender equality in the household,

(Z for respondents with daughters) > (Z for respondents without daughters).

Variables

- **Dependent Variable**

We construct a gender equality indizes as follows:

variable	statement	dataset	cohorts	recoding	measure
pli0043_h	Hausarbeit Std., Werktg.	pl	1984-2022	• only consider positive values • absolute value of the difference between woman and man in household in percent	behavior

Additional coding and transformation

We standardize the direction of responses so that higher values indicate stronger gender equality.

Additionally, in each year, we compute the z-score:

$z_{ij} = (x_{ij} - \mu_j) / \sigma_j$

Interpretation:

- The index X_i has mean 0 and is in standard deviation units.
- Higher values indicate higher gender equality.

- **Independent Variable: Having Daughters**

The number of sons and daughters can be identified using the variables kidsex01-kidsex19 in the biobirth dataset of the SOEP.

These variables record the biological sex of each child, coded as 1 = male and 2 = female. By counting the occurrences of each value for a given respondent or household, one can determine the number of sons, daughters, and kids respectively.

As an explanatory variable, we define "At Least 1 Girl" as a binary indicator that equals 1 if an individual has at least one daughter, and 0 if the individual has exclusively sons.

- **Control Variables**

We include a set of control variables to account for potential confounders:

Concept	Variable Name (Example)	Dataset	Why It's a Key Control	valid obs	recoding
Average Age	bioage	biobirth	Older respondents are more likely to have children and may hold more traditional views.	12398	only consider positive values, take average, take ln(x)
Income	pglabnet	pgen	Net labor income is an essential control variable as income levels can shape access to education, social networks, and exposure to egalitarian values, all of which may affect attitudes toward gender equality.	39877	take sum, take ln(x)
Dummys for the number of children	kidsex01-kidsex19	biobirth	More children → higher chance of having a daughter; family values may vary by family size.	≤9862	19 dummy variables, 1 if the number of children in household matches, otherwise 0
Dummy for migration background	migback	ppathl	Cultural background can influence both fertility and gender role attitudes.	129787	1 if migback=2 (direct migration background) or 3 (indirect migration background) for at least one of the respondents, 0 otherwise
Year of observation	syear	pl			Year fixed effects
Household	pid	pl			Household fixed effects

2. Inclusion/Exclusion Criteria

Include:

- Respondents who participated in the individual questionnaire (pl) in any year from 1984 to 2022.

Exclude:

- Respondents without any biological children
- Respondents with missing values for the key controls
- Respondents with unmatched IDs (pid) across datasets
- Respondents who cannot be assigned to a household in pairs

3. Exact analysis

Panel Approach (with Fixed Effects): OLS regression with robust standard errors (clustered at person level for panel specifications). Panel model:

$X_it = \beta_0 + \beta_1 \cdot$ *At Least 1 Girl*$_it + \gamma' Z_it + \alpha_i + \delta_t + \varepsilon_it$

Where:

- X = gender attitudes index
- "At Least 1 Girl" = indicator from biobirth
- Z = controls from biobirth, ppathl
- δ_t = year fixed effects
- α_i = household fixed effects (if panel)

4. Interpretation of Results

- Focus: Sign and statistical significance of β_1
- Report:
  - Coefficient estimates
  - Robust standard errors
  - t-values and p-values
  - 95% confidence intervals
- Hypothesis is supported if $\beta_1 > 0$ and statistically significant ($p < 0.05$)

Hypothesis

Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes). That is, denoting X as a measure of support for gender equality,

(X for male respondents with daughters

− X for male respondents without daughters)

(X for female respondents with daughters

− X for female respondents without daughters)

1. Variables

- **Dependent Variable (Outcome)**

We construct a gender attitudes index as follows:

variable	statement	dataset		cohorts	valid obs	recoding
plj0625	At home, the husband should have the final say.	pl	bgp/bgpr389 bhp/bhp_508_q56 bip/bip_471_q107	2016 2017 2018	4166 2827 420	only consider values from 1 to 7
plj0624	For parents, vocational training or higher education for their sons should be more important than vocational training or higher education for their daughters.	pl	bgp/bgpr388 bhp/bhp_507_q56 bip/bip_470_q107	2016 2017 2018	4117 2803 416	only consider values from 1 to 7
plj0621	Having a job is the best way for a woman to be independent.	pl	bgp/bgpr385 bhp/bhp_504_q56 bip/bip_467_q107	2016 2017 2018	4105 2793 415	only consider values from 1 to 7

plj0622	Even a married woman should have a paid job so that she can be financially independent.	pl	bgp/bgpr386 bhp/bhp_505_q56 bip/bip_468_q107	2016 2017 2018	4099 2796 417	only consider values from 1 to 7
plj0623	If a woman earns more money than her partner, this inevitably leads to problems.	pl	bgp/bgpr387 bhp/bhp_506_q56 bip/bip_469_q107	2016 2017 2018	3983 2745 406	only consider values from 1 to 7

Additional coding and transformation
To ensure consistency across all survey items, we standardize the direction of responses so that higher values always indicate stronger support for gender equality.

Construction of the Standardized Average Index
Let x_{ij} be the recoded response of respondent i to item j. Let μ_j and σ_j be the mean and standard deviation of item j across all respondents.

1. Standardization:
For each item j and year, compute the z-score:
$z_{ij} = (x_{ij} - \mu_j) / \sigma_j$

2. Index construction:
Compute the mean of the standardized responses for each individual:
$X_i = (1 / n_i) * \sum z_{ij}$ over j in J_i, where J_i is the set of items answered by respondent i.

3. Interpretation:
- The index X_i has mean 0 and is in standard deviation units.
- Higher values indicate more egalitarian attitudes.

- **Independent Variable: Having Daughters**

The number of sons and daughters can be identified using the variables kidsex01-kidsex19 in the biobirth dataset of the SOEP.

These variables record the biological sex of each child, coded as 1 = male and 2 = female. By counting the occurrences of each value for a given respondent or household, one can determine the number of sons, daughters, and kids respectively.

As an explanatory variable, we define "At Least 1 Girl" as a binary indicator that equals 1 if an individual has at least one daughter, and 0 if the individual has exclusively sons.

- **Control Variables**

We include a set of control variables to account for potential confounders:

Concept	Variable Name (Example)	Dataset	Why It's a Key Control	valid obs	recoding
Age	bioage	biobirth	Older respondents are more likely to have children and may hold more traditional views.	12398	only consider positive values, take ln(x)
Dummy for gender (respondent)	sex	biobirth	Gender affects both attitudes and fertility patterns.	24317	1 if sex=2 (female), 0 otherwise
Dummys for the number of children	kidsex01-kidsex19	biobirth	More children → higher chance of having a daughter; family values may vary by family size.	≤9862	19 dummy variables, 1 if the number of children matches, otherwise 0
Dummys for martial status	pgfamstd	pgen	Being married or in a partnership may influence both the likelihood of having children (including daughters) and one's social and political attitudes, including views on gender equality.	74742	4 dummy variables for "married", "married but separated", "divorced", "widowed", 1 if applicable, 0 otherwise
Current net labor income	pglabnet	pgen	Net labor income is an essential control variable as income levels can shape access to education, social networks, and exposure to egalitarian values, all of which may affect attitudes toward gender equality.	39877	only consider positive values, take ln(x)
Dummy for migration background	migback	ppathl	Cultural background can influence both fertility and gender role attitudes.	129787	1 if migback=2 (direct migration background) or 3 (indirect migration background), 0 otherwise

Year of observation	syear	pl	Year fixed effects		Year fixed effects
Never Changing Person ID	pid	pl	Individual fixed effects		Individual fixed effects

2. Inclusion/Exclusion Criteria

Include:

- Respondents who participated in the individual questionnaire (pl) in any year from 2016 to 2018.

Exclude:

- Respondents without any biological children
- Respondents with fewer than 2 answered gender attitude variables
- Respondents with missing values for the key controls
- Respondents with unmatched IDs (pid) across datasets

3. Exact analysis

Panel Approach (with Fixed Effects): OLS regression with gender interaction to test "larger increase in the support of gender equality for men than for woman;" robust standard errors (clustered at person level for panel specifications). Panel model:

$X_it = \beta_o + \beta_1 \cdot$ *At Least 1 Girl* $_it + \beta_2 \cdot$ *At Least 1 Girl* $_it * sex_i + \gamma' Z_it + \alpha_i + \delta_t + \varepsilon_it$

Where:

- X = gender attitudes index
- "At Least 1 Girl" = indicator from biobirth
- sex = gender (male = 1, female = 0) of respondent
- Z = controls from biobirth, pgen, ppathl
- δ_t = year fixed effects
- α_i = person fixed effects (if panel)

4. Interpretation of Results

- Focus: Sign and statistical significance of β_1
- Report:
  - Coefficient estimates
  - Robust standard errors
  - t-values and p-values
  - 95% confidence intervals
- Hypothesis is supported if $\beta_1 > 0$ and statistically significant ($p < 0.05$)